

Results

目前导出为pdf时排版有点混乱

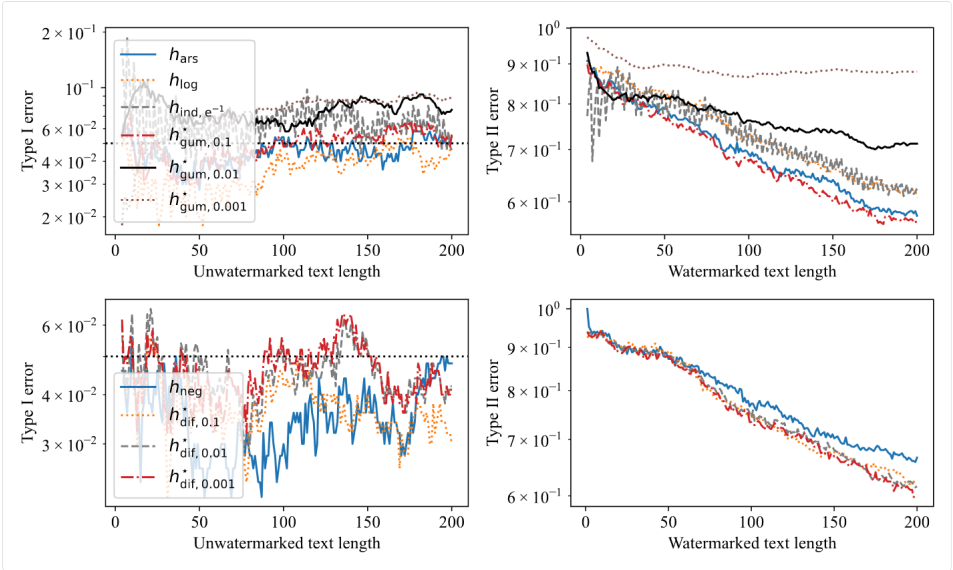
1p3B stands for facebook/opt-1.3b
2p7B stands for princeton-nlp/Sheared-LLaMA-2.7B
7B stands for huggyllama/llama-7b.

C4 news

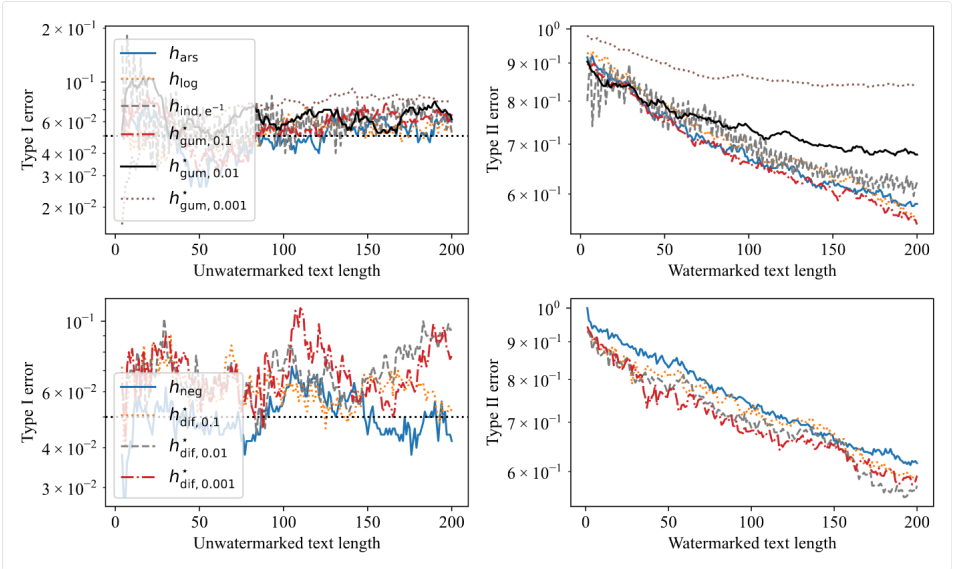
Comparison 1. Only change the model

All other arguments remain the same:
--c 4 --m 200 --T 500 --seed_way noncomm_prf --seed 111 --temp 0.1

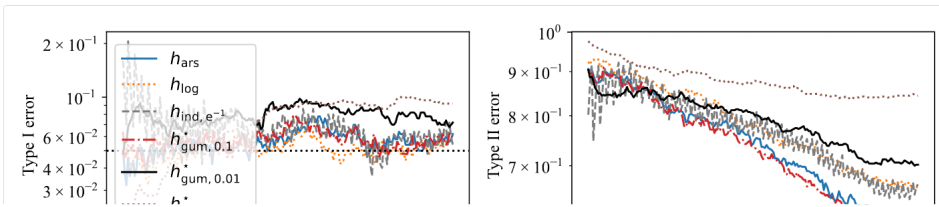
1p3B



2p7B



7B



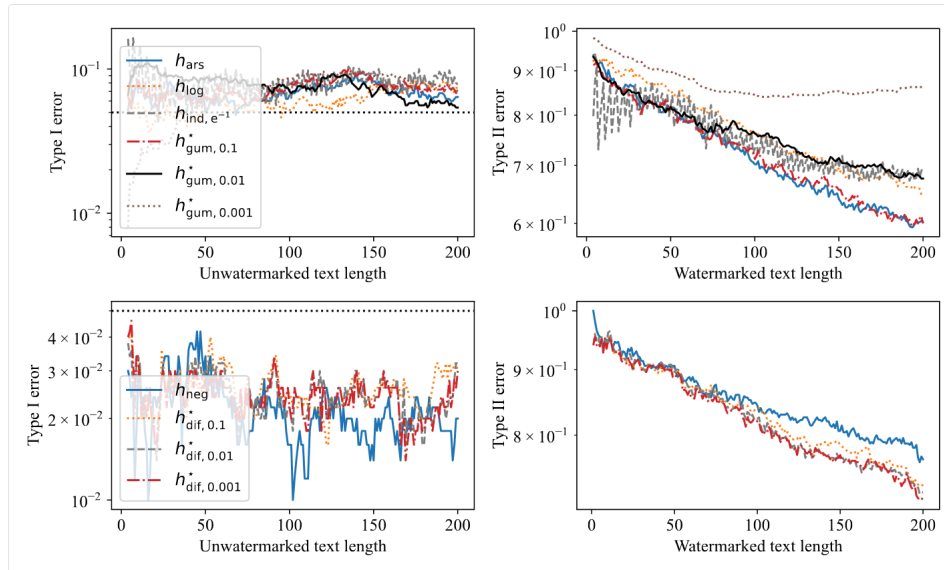
Their results are kind of similar. Our initial guess is that the choice of model does not have a significant impact on type 1 and type 2 error.

Comparison 2. Choice of hash function: skipgram and noncomm

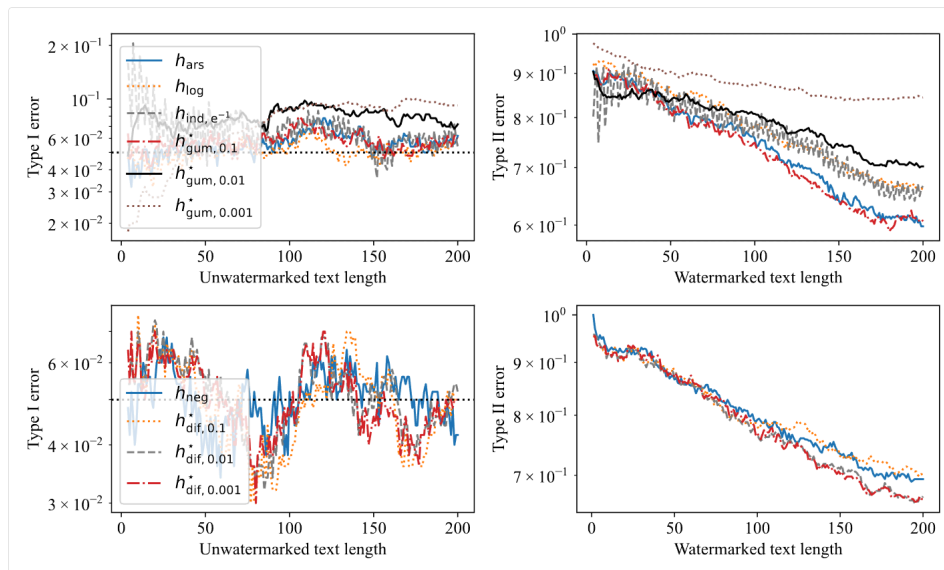
All other arguments remain the same:

7B --c 4 --m 200 --T 500 --seed 111 --temp 0.1

skipgram



noncomm



We see the differences lies in the type 1 errors. It has more impacts on inverse transform than gumbel. Why the type 1 error of inverse transform is low for skipgram_prf? In theory it is because the "c" for skipgram represents "skip distance".

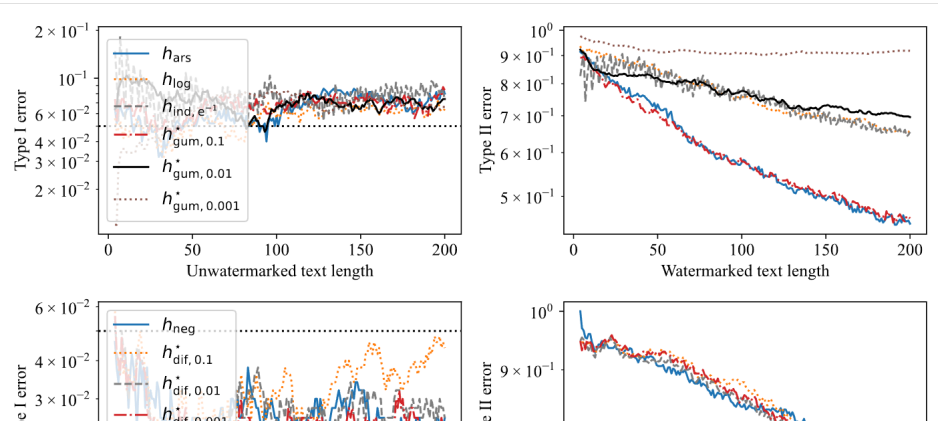
Say $c = 4$, and suppose the 4 most recent tokens are $[a, b, c, d]$, then the seed is only determined by a , not a, b, c, d jointly. This is to ensure robustness, but will cause hash collisions. And so the iid assumption could not be met.

Comparison 3. skipgram: $c = 1$ and $c = 4$

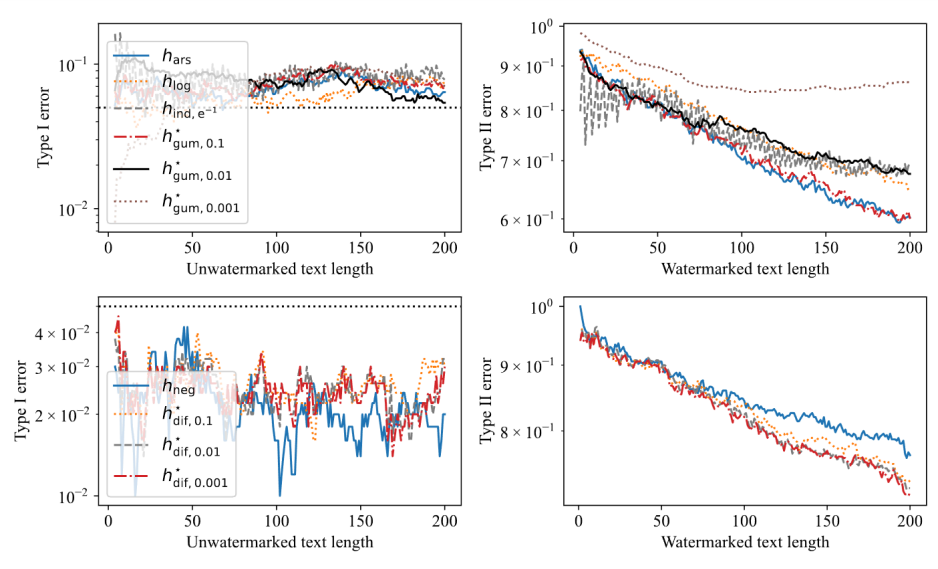
All other arguments remain the same:

7B --m 400 --T 1000 --seed_way skipgram_prf --seed 111 --temp 0.1

c=1



$c=4$



They are mostly identical, except for gumbel, h_{ars} and $h_{gum, 0.1}^*$ decay a little faster in the $c = 1$ case.

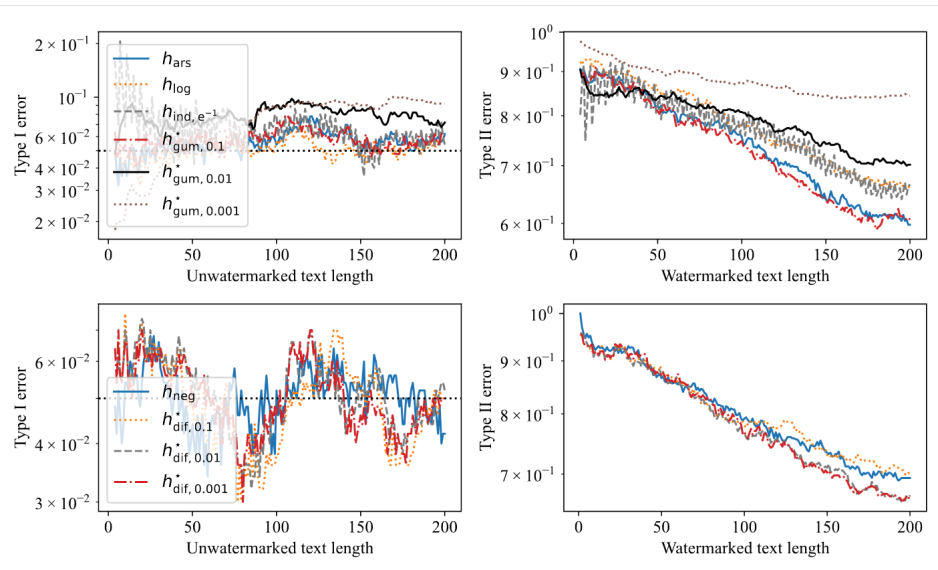
Comparison 4. Temperatures

Control group 1

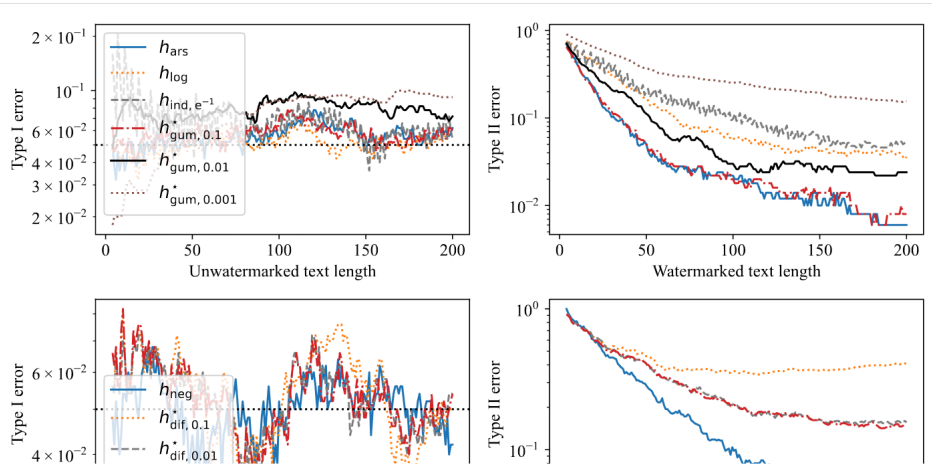
All other arguments remain the same:

7B --c 4 --m 200 --T 500 --seed_way noncomm_prf --seed 111

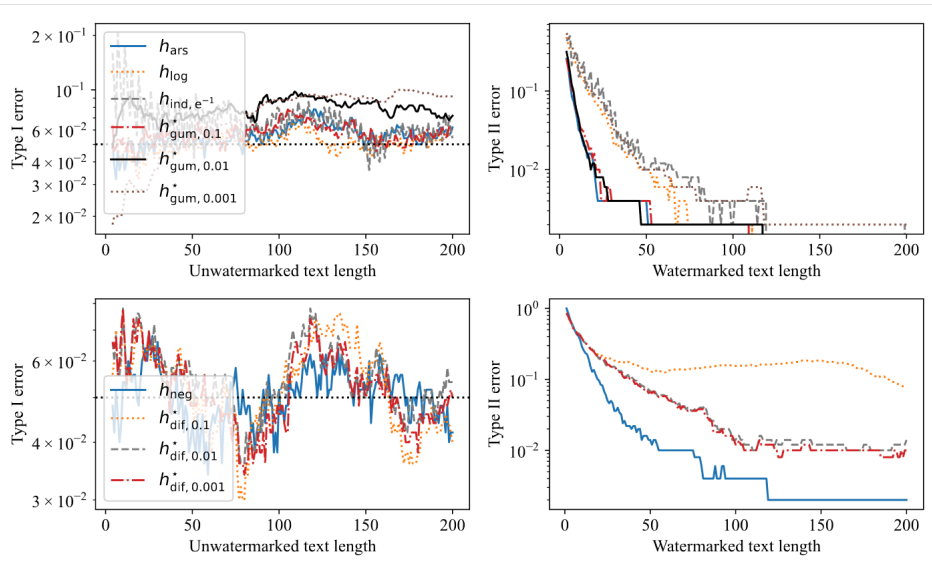
$T=0.1$



$T=0.5$



T=1.0



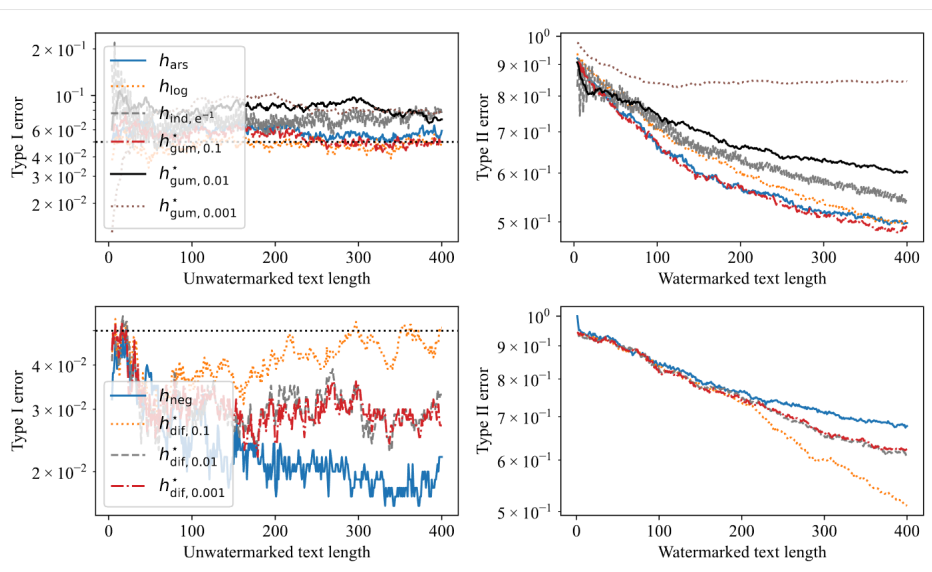
Type 1 errors are identical, which aligns with theory. Larger T corresponds to better type 2 error control as expected.

Control group 2

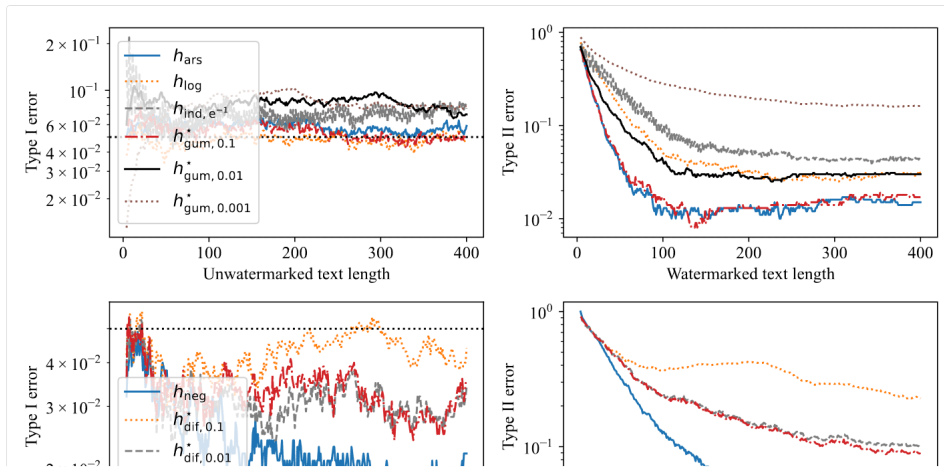
All other arguments remain the same:

7B --c 4 --m 400 --T 1000 --seed_way skipgram_prf --seed 111

T=0.1



T=0.5



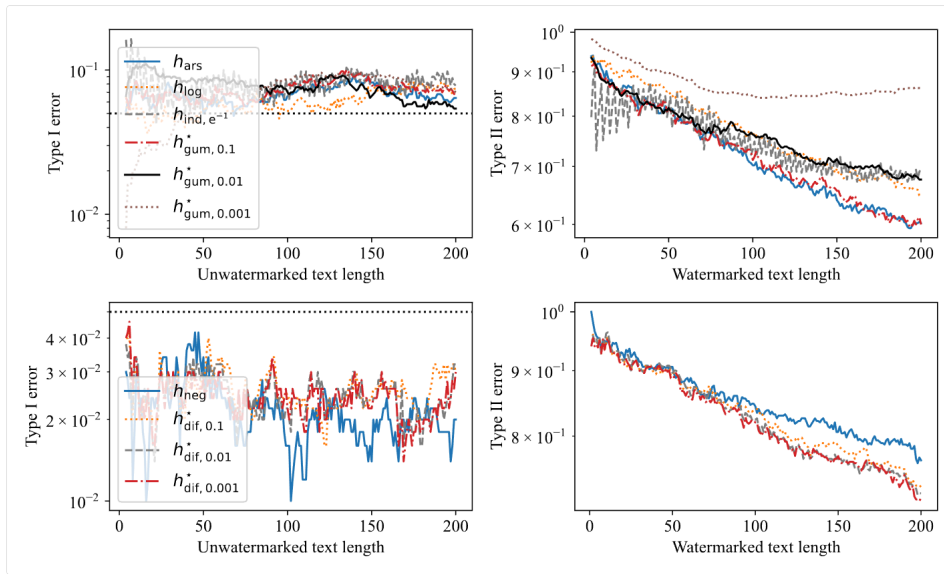
We get similar conclusions. Note that the rank of the curves have changed (changing temperature might result in different efficiency class).

Comparison 4. length of generated tokens and number of iteration

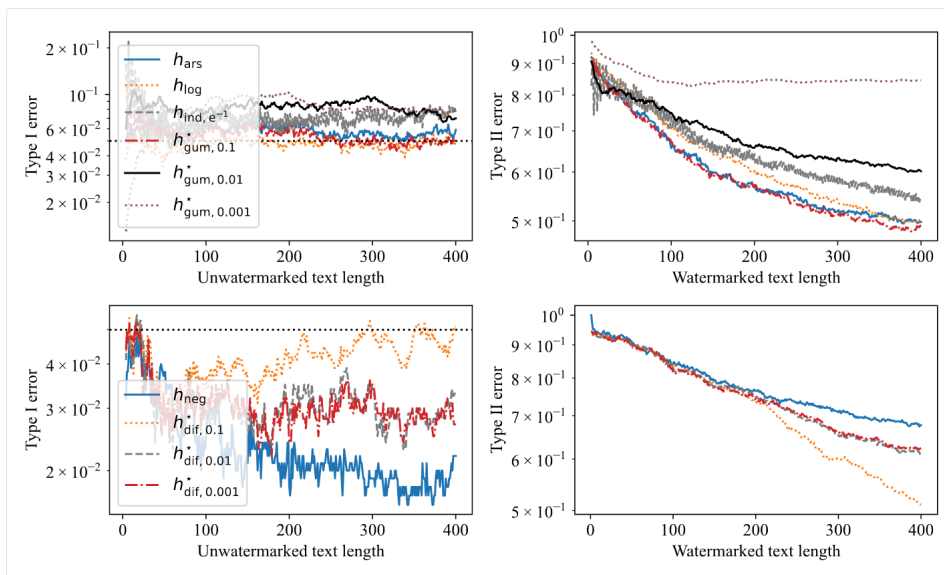
All other arguments remain the same:

7B --c 4 --seed_way skipgram --seed 111 --temp 0.1

m=200, T=500



m=400, T=1000



We see larger m and T gives slightly better type 2 error control (temp is set to 0.1, which corresponds to a "worse case scenario").

ChatDoctor

Comparison 1. C4News vs. ChatDoctor

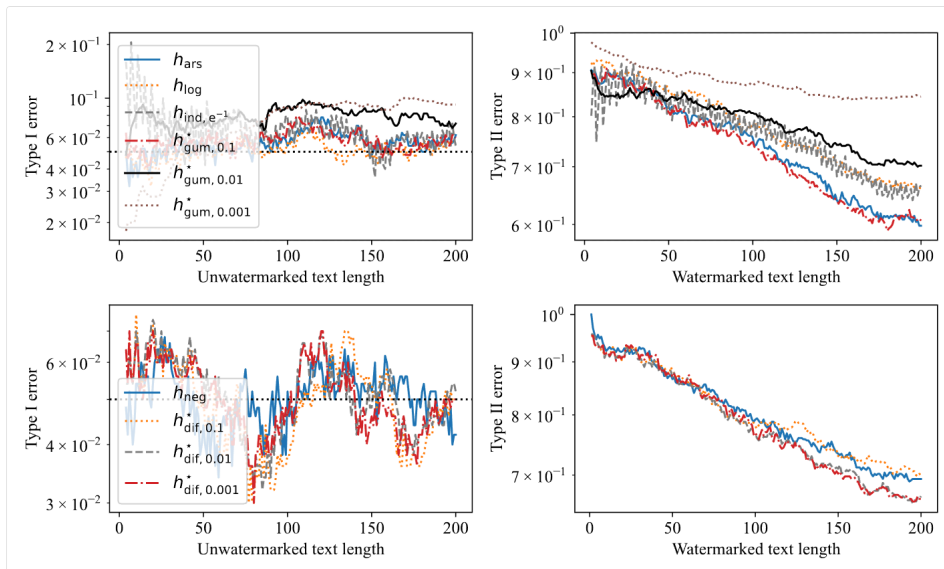
Control group 1

All other arguments remain the same:

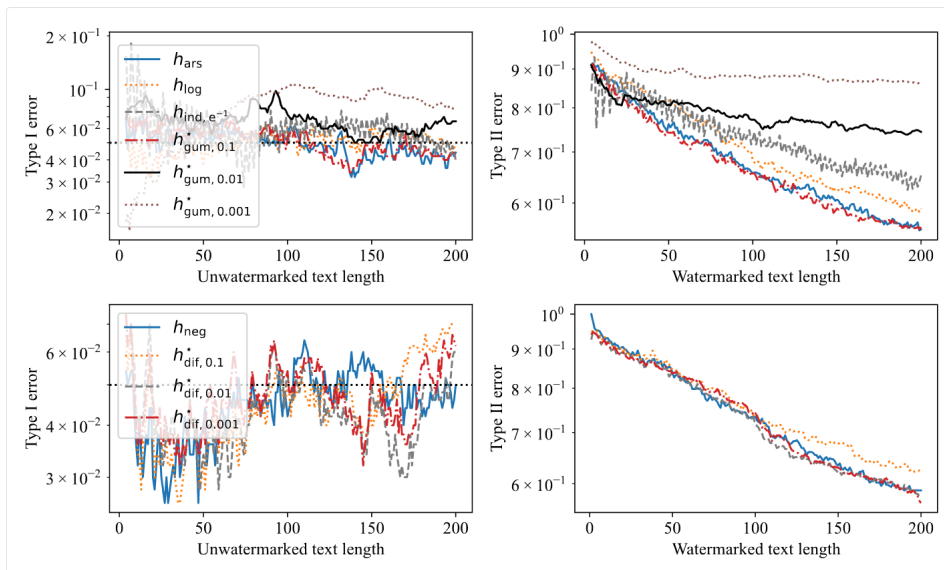
7B --c 4 --m 200 --T 500 --seed_way noncomm_prf --seed 111 --temp 0.1

We only change dataset.

C4News



ChatDoctor



Temp is set to 0.1. For gumbel type 2 errors, the results are similar. For transform type 2 errors, ChatDoctor has slightly better control.

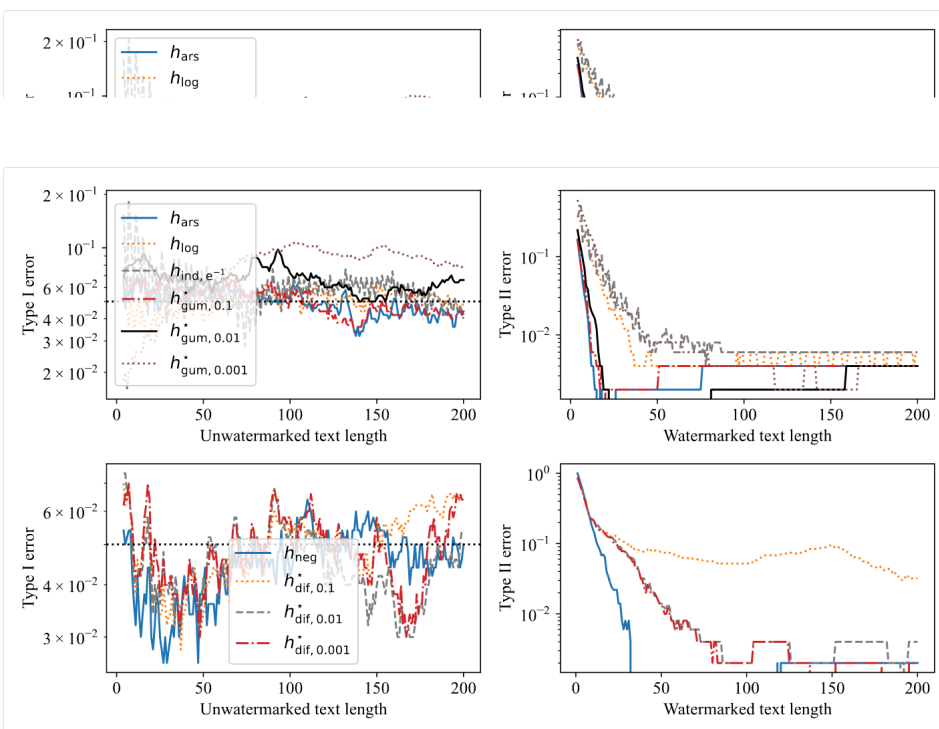
Control group 2

All other arguments remain the same:

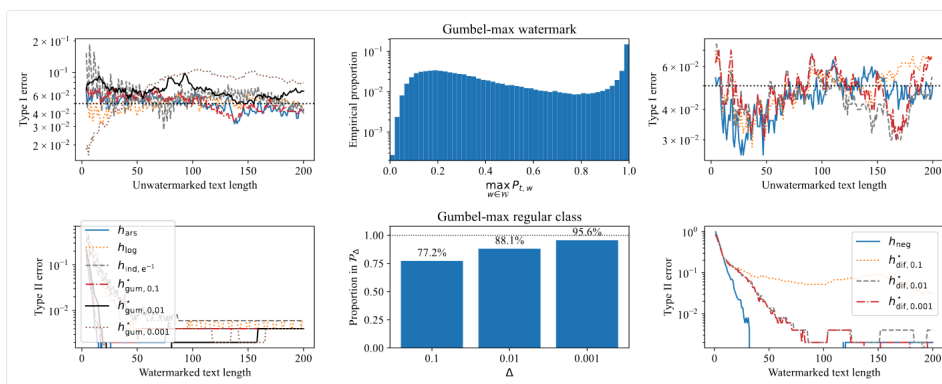
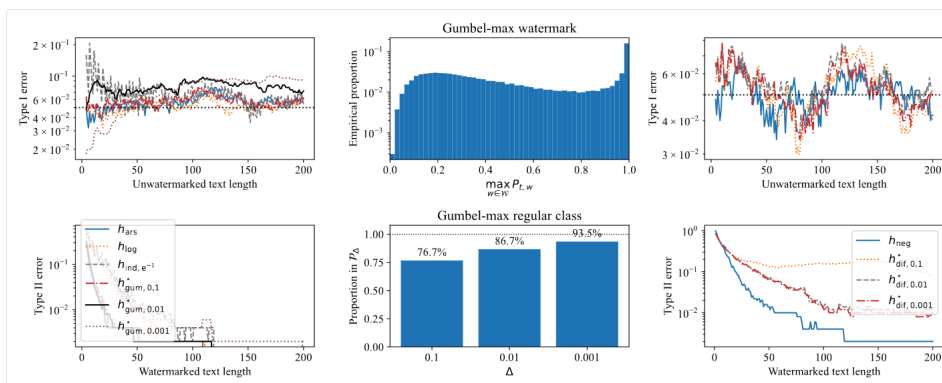
7B --c 4 --m 200 --T 500 --seed_way noncomm_prf --seed 111 --temp 1

C4News

ChatDoctor



Temp is set to 1, which is a more realistic scenario.

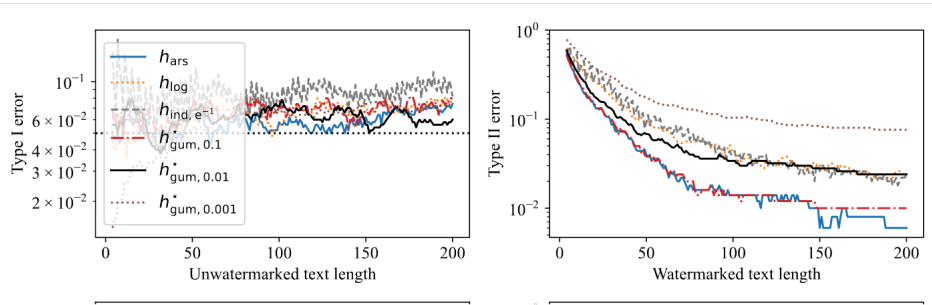


StackMathQA

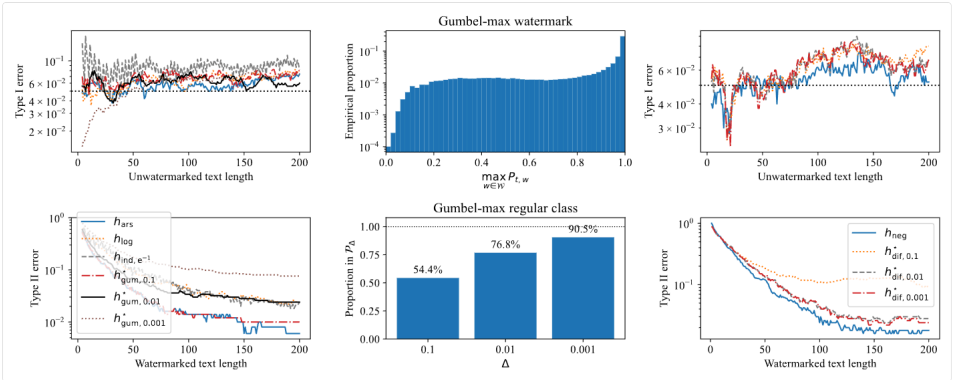
All other arguments remain the same:

7B --c 4 --m 200 --T 500 --seed_way noncomm_prf --seed 111 --temp 1

StackMathQA



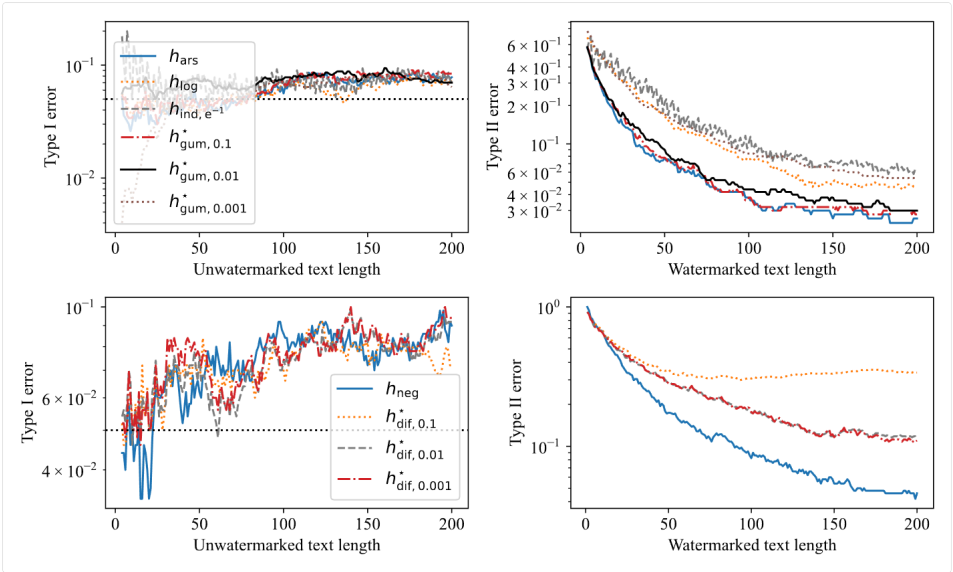
Temp is set to 1, which is a more realistic scenario.



Codeparrot

All other arguments remain the same:
 7B --c 4 --m 200 --T 500 --seed_way noncomm_prf --seed 111 --temp 1

Codeparrot



Temp is set to 1, which is a more realistic scenario.

